Electronics 2 Project Design and Evaluation Report (Phase Two)

Full name: Soheil Bolvardi **Student ID:** 403105811

1 Required Project Specifications

Main Requirements of Phase Two

In this phase, the completed amplifier circuit must be designed and evaluated to include a differential input, voltage amplification stages, bias circuits, a power output stage, an output load, and global negative feedback. The final circuit must meet the following specifications:

Value	Parameter
50Ω	Output load resistance (R_L)
$18 \leq \lvert A_{v,\text{closed}} \rvert \leq 22$	Closed-loop gain at 1 kHz
$10k\Omega$ to $200k\Omega$	Permitted range of resistances R_f and R_g
$16V_{pp}$	Main output swing without clipping
$> 60\%$	Output-stage efficiency at a swing of $16V_{pp}$
$< 0.08\%$	THD under normal conditions
$< 1\%$	THD in the presence of a noise source
$\leq 190mW$	Total power consumption for a $50mV$ input at 1 kHz
$> 1M\Omega$	Differential input resistance ($R_{in,\text{diff}}$)
$< 50\Omega$	Output resistance (R_{out}) without R_L
$10mV$	Sawtooth supply ripple in the $PSRR$ test
≤ 200	Permitted circuit cost

Final Specification Verification Table

At the end of the report, enter your final results in the table below.

Parameter	Required value	Designed value
$\lvert A_{v,\text{closed}} \rvert$	18 to 22	19.99
$V_{out,pp}$	$16V_{pp}$	$16.97V_{pp}$
η_{out}	$> 60\%$	72.06%
P_{total}	$\leq 190mW$	$175.48mV$
Normal THD	$< 0.08\%$	0.022%
THD with a noise source	$< 1\%$	0.463%
$PSRR$	—	$87.83dB$
$R_{in,\text{diff}}$	$> 1M\Omega$	∞
R_{out}	$< 50\Omega$	$4.90m\Omega$
Cost	≤ 200	109

2 Introduction to the Final Circuit and Overall Structure

Problem Requirements (Section 1)

Introduce the final project circuit. Emphasize that only one complete final circuit has been submitted and that this circuit includes the following sections:

- Differential input
- Voltage amplification stages
- Bias circuits
- Power output stage
- Output load $R_L = 50\Omega$
- Global negative feedback network

Also explain the sequence in which the circuit was completed: first adding the output stage and then applying feedback.

Introduction to the Selected Final Structure

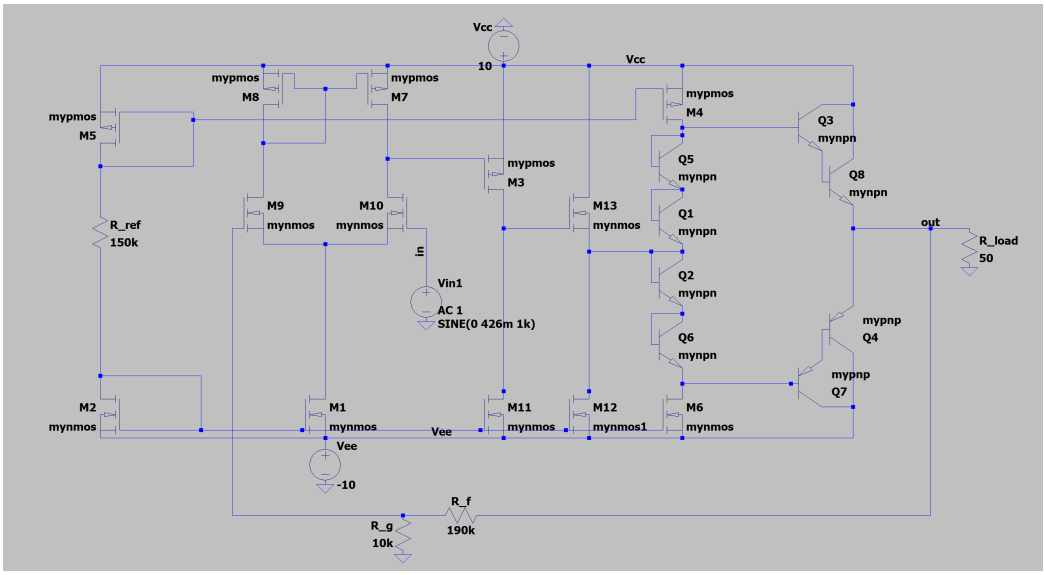
In this section, explain the overall structure of your circuit. For example:

- What is the structure of your output stage?
- From which node is feedback taken, and to which input is it returned?
- What changes have been made relative to the phase-one circuit? (If the circuit from the previous phase was not used at all, briefly explain why it was not used and what its problems were.)

In summary, the circuit consists of a differential input stage, a voltage amplification stage, a voltage buffer stage (which prevents the output stage from affecting the bias of the preceding stages), and finally a Push Pull Class AB stage. Feedback is taken from the output node and connected to the differential input. The overall circuit structure is similar to that of phase one. However, the bias currents have been significantly reduced (by increasing the reference resistance) to reduce the circuit's operating power. In addition, the BJT transistors in the differential and buffer stages have been replaced with MOS transistors, and an amplifier stage has also been added to the circuit structure, which was not present in phase one.

Complete Schematic of the Final Circuit

The complete schematic of the final circuit is shown below:


Vcc
10
Vcc
mypmos
M8
mypmos
M7
mypmos
M5
mypmos
M4
Q3
mynpn
mypmos
M3
Q5
mynpn
Q8
mynpn
M9
mynmos
M10
mynmos
M13
mynmos
Q1
mynpn
out
R_load
50
R_ref
150k
in
Vin1
AC 1
SINE(0 426m 1k)
Q2
mynpn
mypnp
Q4
Q6
mynpn
mypnp
Q7
M2
mynmos
M1
mynmos
M11
mynmos
M12
mynmos1
M6
mynmos
Vee
Vee
-10
R_f
190k
R_g
10k

3 DC and Bias Analysis of the Final Circuit

Problem Requirements (Section 2)

Perform a DC analysis of the final circuit. Examine the operating points of the main stages, the bias currents, the voltages at important nodes, and the bias condition of the output stage. If you have used a V_{BE} multiplier or a similar circuit, also report its bias value.

Theoretical Bias Calculations for the Final Circuit

The manual calculations for:

- The main circuit currents
- The voltages at important nodes
- The output-stage bias

should be included in this section.

* جریان بایاس شاخه مرجع:

$$I_{ref} = \frac{20 - V_{GS,NMOS} - V_{SG,PMOS}}{R_{ref}} ,$$

$R_{ref} = 150\,k\Omega$, $V_{GS,NMOS} = V_{DS,NMOS}$, $V_{SG,PMOS} = V_{DS,PMOS}$

$$\rightsquigarrow \begin{cases} NMOS: \frac{20 - V_{GS,NMOS} - V_{SG,PMOS}}{150} = \frac{1}{2} \times 0.25 \times 100 \times (V_{GS,NMOS} - 0.6)(1 + 0.05 V_{GS,NMOS}) \\ PMOS: \frac{20 - V_{GS,NMOS} - V_{SG,PMOS}}{150} = \frac{1}{2} \times 0.08 \times 50 \times (V_{SG,PMOS} - 0.6)(1 + 0.05 V_{SG,PMOS}) \end{cases}$$

← از حل دستگاه بالا داریم:

$V_{GS,NMOS} \simeq 0.7\,V$, $V_{SG,PMOS} \simeq 0.85$

$$\rightsquigarrow I_{ref} \simeq 123\,\mu A = I_{M2} = I_{M5}$$

* طبقه تفاضلی

$V_{in,DC} = 0 \rightsquigarrow V_{DS,M_1} = -V_{GS,M_{10}} + 10$, $V_{GS_1} = 0.7\,V$

شبه بخش قبل: $V_{SG,M7} = V_{SG,M8} \simeq 0.85\,V = V_{Drain,M10} \rightsquigarrow V_{DS,10} \simeq 9.15 + V_{GS,10}$

$$\begin{cases} I_{D,M_1} = \frac{1}{2} \times 0.25 \times 100 \times (0.7 - 0.6)^2 (1 + 0.05(10 - V_{GS,M_{10}})) \\ I_{D,M10} = \frac{1}{2} I_{D,M_1} = \frac{1}{2} \times 0.25 \times 100 \times (V_{GS,M10} - 0.6)^2 (1 + 0.05(9.15 + V_{GS,M10})) \end{cases}$$

← از دستگاه بالا:

$$I_{D,M_1} \simeq 175\,\mu A$$

$$\rightsquigarrow I_{D,M9} = I_{D,M10} \simeq 85.75\,\mu A$$

← طبقه تقویت کننده Common - Source: $I_{D,M11} = I_{D,M3} = I_{D,M1} \simeq 175\,\mu A$

← طبقه بافر: $\left(\frac{W}{L}\right)_{M,12} = 4\left(\frac{W}{L}\right)_{M,1} \rightsquigarrow I_{D,M12} = I_{D,M13} = 4 I_{DM_1} \simeq 0.7\,mA$

← طبقه خروجی:

$$I_{D,M4} \simeq I_{D,M6} = I_{DM_1} \simeq 175\,\mu A$$

طبق بخش قبل : $V_{SG,M8} \simeq 0.85^V = V_{DS,M8} \rightarrow V_{DS,M8} = 0.85^V$

ولتاژ خروجی طبقه دیفرانسیلی $\simeq 10 - 0.85 = 9.15^V$

$V_{tail} = -V_{GS,M10} \simeq -0.7^V$ (بخش قبل)

$I_{D,M3} \simeq 175^{mA}$, $V_{SG,M3} = 9.15 - V_{D,M3}$, $V_{SD,M3} = 10 - V_{D,M3}$

$\rightsquigarrow 0.175 = \frac{1}{2} \times 0.08 \times 50 \times (9.15 - V_{D,M3} - 0.6)^2 (1 + 0.05(10 - V_{D,M3}))$

ولتاژ خروجی طبقه CS : $V_{D,M3} \simeq 0.85^V$

به طور مشابه : $I_{D,M13} \simeq 700^{mA}$, $V_{GS,M13} \simeq 0.85 - V_{S,M13}$, $V_{DS,M13} = 10 - V_{S,M13}$

$\rightsquigarrow 0.7 = \frac{1}{2} \times 0.25 \times 0.8 \times (0.85 - V_{S,M13} - 0.6)^2 (1 + 0.05(10 - V_{S,M13}))$

ولتاژ خروجی بافر : $V_{S,M13} \simeq -0.08^V$

با توجه به کوچک بودن ولتاژ وارد شده به طبقه خروجی و همچنین استفاده از طبقه Class AB Push-Pull انتظار داریم:

$V_{out} \simeq 0^V$

DC Operating Point Simulation Results

In this section, include the output of the DC operating point simulation and extract the important values.

Ig(M3):	-0	device_current
Is(M3):	0.000183062	device_current
Ib(M3):	9.22144e-12	device_current
Id(M8):	-8.71039e-05	device_current
Ig(M8):	-0	device_current
Is(M8):	8.71039e-05	device_current
Ib(M8):	8.14616e-13	device_current
Id(M12):	0.000713337	device_current
Ig(M12):	0	device_current
Is(M12):	-0.000713337	device_current
Ib(M12):	-1.00034e-11	device_current
Id(M13):	0.000714409	device_current
Ig(M13):	0	device_current
Is(M13):	-0.000714409	device_current
Ib(M13):	-1.00166e-11	device_current
I(R_ref):	0.000123063	device_current
Ic(Q2):	0.000170142	device_current
Ib(Q2):	8.51073e-07	device_current
Ie(Q2):	-0.000170993	device_current
Is(Q2):	0	device_current
Id(M4):	-0.000169974	device_current
Ig(M4):	-0	device_current
Is(M4):	0.000169974	device_current
Ib(M4):	8.79824e-12	device_current
Id(M7):	-8.72953e-05	device_current
Ig(M7):	-0	device_current
Is(M7):	8.72953e-05	device_current
Ib(M7):	8.60336e-13	device_current
Id(M10):	8.72953e-05	device_current
Ig(M10):	0	device_current
Is(M10):	-8.72953e-05	device_current
Ib(M10):	-9.82811e-12	device_current
Ic(Q8):	0.0024927	device_current
Ib(Q8):	1.14653e-05	device_current
Ie(Q8):	-0.00250417	device_current
Is(Q8):	0	device_current
Id(M11):	0.000183062	device_current
Ig(M11):	0	device_current
Is(M11):	-0.000183062	device_current
Ib(M11):	-1.07986e-11	device_current
Id(M2):	0.000123063	device_current
Ig(M2):	0	device_current
Is(M2):	-0.000123063	device_current
Ib(M2):	-7.07536e-13	device_current
Ic(Q4):	-0.00253957	device_current
Ib(Q4):	-1.55771e-05	device_current
Ie(Q4):	0.00255515	device_current
Is(Q4):	0	device_current
I(R_load):	-5.09725e-05	device_current
Id(M9):	8.71039e-05	device_current
Ig(M9):	0	device_current
Is(M9):	-8.71039e-05	device_current
Ib(M9):	-9.87382e-12	device_current
Ic(Q6):	0.000170142	device_current
Ib(Q6):	8.51073e-07	device_current
Ie(Q6):	-0.000170993	device_current
Is(Q6):	0	device_current
I(R_f):	-1.27431e-08	device_current
I(Vcc):	-0.00386902	device_current

--- Operating Point ---

V(n002):	9.15701	voltage
V(n009):	-0.668441	voltage
V(in):	0	voltage
V(n010):	-0.00661847	voltage
V(n003):	1.21176	voltage
V(n012):	-0.680186	voltage
V(out):	-0.00254862	voltage
V(n004):	9.14966	voltage
V(n007):	-0.000127431	voltage
V(vee):	-10	voltage
V(n013):	-9.30246	voltage
V(n006):	0.602573	voltage
V(n011):	-0.615972	voltage
V(vcc):	10	voltage
V(n005):	0.674515	voltage
V(n008):	0.788557	voltage
V(n014):	-1.22533	voltage
V(n001):	9.19538	voltage
Id(M1):	0.000174399	device_current
Ig(M1):	0	device_current
Is(M1):	-0.000174399	device_current
Ib(M1):	-9.34156e-12	device_current
I(Vin1):	0	device_current
Ic(Q1):	0.000169076	device_current
Ib(Q1):	8.45738e-07	device_current
Ie(Q1):	-0.000169922	device_current
Is(Q1):	0	device_current
Id(M5):	-0.000123063	device_current
Ig(M5):	-0	device_current
Is(M5):	0.000123063	device_current
Ib(M5):	8.52988e-13	device_current
Id(M6):	0.000171088	device_current
Ig(M6):	0	device_current
Is(M6):	-0.000171088	device_current
Ib(M6):	-8.78467e-12	device_current
Ic(Q5):	0.000169076	device_current
Ib(Q5):	8.45738e-07	device_current
Ie(Q5):	-0.000169922	device_current
Is(Q5):	0	device_current
Ic(Q7):	-1.54822e-05	device_current
Ib(Q7):	-9.48834e-08	device_current
Ie(Q7):	1.55771e-05	device_current
Is(Q7):	0	device_current
I(R_g):	1.27431e-08	device_current
I(Vee):	0.00392001	device_current
Ic(Q3):	1.14129e-05	device_current
Ib(Q3):	5.24477e-08	device_current
Ie(Q3):	-1.14653e-05	device_current
Is(Q3):	0	device_current
Id(M3):	-0.000183062	device_current
Ig(M3):	-0	device_current
Is(M3):	0.000183062	device_current
Ib(M3):	9.22144e-12	device_current
Id(M8):	-8.71039e-05	device_current
Ig(M8):	-0	device_current
Is(M8):	8.71039e-05	device_current
Ib(M8):	8.14616e-13	device_current

Key results:

- DC output voltage with zero input: $-2.55\,mV$
- Output-stage bias current: $171\,\mu A$
- Brief explanation of the bias condition and its stability: Because of the current mirrors in the differential and output stages, these stages have no stability issues. Given the use of active loads in the common-source and buffer stages, if the circuit is to be built using discrete components, there will be a dependence on transistor area. However, the open-loop gain will still be sufficiently large. The buffer stage also prevents the output stage from changing the bias of the preceding stages, making the circuit more stable.

4 Power Output Stage Design

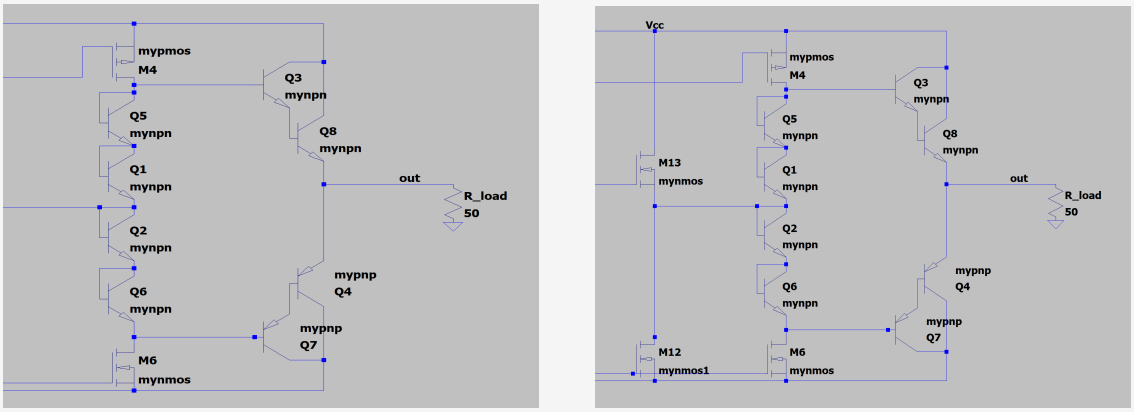
Problem Requirements (Section 3)

Introduce and analyze the power output stage. You must show that this stage is integrated into the main circuit and is designed to drive a 50Ω load. In this section, explain the connections, transistor types, biasing, and reasons for selecting this structure.

Theoretical Analysis and Calculations of the Output Stage

In this section, examine the following:

- The type of output structure and the reason for selecting it
- An approximate calculation of the current required by the load
- The output-stage efficiency under ideal conditions



Right figure: the output stage used - Left figure: the connection of the output stage to the preceding stages through a voltage buffer.

The theoretical calculations for this output stage are provided in the image below:

★ برای طبقه خروجی، از ساختار Push-Pull Class AB استفاده شده است که ترانزیستورهای قدرت آن به صورت زوج دارلینگتون هستند.

دلیل: با توجه به وجود ترانزیستورهای diode-connected، اندازه ناحیه مرده تقریبا صفر است بنابراین اعوجاجی در خروجی (هنگام عبور از صفر) دیده نخواهد شد. همچنین این ساختار zizo است. علت استفاده از زوج دارلینگتون هم تأمین جریان بار است.

zizo

{ نیم‌سیکل مثبت: Q_3, Q_8: ON, Q_4, Q_7: OFF $\rightarrow v_{out} = v_{in} + 2v_{BE} - 2v_{BE} = v_{in}$

نیم‌سیکل منفی: Q_3, Q_8: OFF, Q_4, Q_7: ON $\rightarrow v_{out} = v_{in} + 2v_{BE} - 2v_{BE} = v_{in}$

با توجه به اینکه بیشینه سوینگ Peak to peak مدنظر 17 V می‌باشد، بیشینه جریان مورد نیاز بار:

$$I_{load,max} = \frac{V_{p,max}}{50} = \frac{8.5}{50} = 170\ mA$$

در صورتی که از زوج دارلینگتون استفاده نمی‌کردیم، بیشینه جریانی که می‌شد تأمین کرد:

$I_{bias} \simeq 170\ mA$

{ نیم‌سیکل مثبت: $I_{l,max} = I_{bias} \times \beta_{npn} = 34\ mA$

نیم‌سیکل منفی: $I_{l,max} = I_{bias} \times \beta_{pnp} = 25.5\ mA$

که می‌بینیم بسیار کمتر از جریان مورد نیاز هستند. به همین دلیل از زوج دارلینگتون استفاده شده است. در این صورت:

{ نیم‌سیکل مثبت: $I_{l,max} = I_{bias} \times \beta_{npn}^2 = 6.8\ A$ ✓

نیم‌سیکل منفی: $I_{l,max} = I_{bias} \times \beta_{pnp}^2 = 3.825\ A$ ✓

→ که این دو مقدار بسیار بیشتر از بیشینه جریان مورد نیاز بار هستند و جریان بار به سادگی تأمین می‌شود.

همچنین از رابطه ساختار Push-Pull در حالت ایده‌آل می‌دانیم:

$$P_{supply} = 2V_{cc}\left(\frac{V_p}{\pi R_L} + I_{bias} + \underbrace{I_{DC,\text{بقیه‌ها}}}_{I_{D,M2} + I_{D,M1} + I_{D,M11} + I_{D,M12}}\right)$$

به دلیل کوپلینگ بودن، از توان AC سایر طبقات صرف نظر کرده‌ایم.

$$\rightarrow P_{supply} \simeq 1.147\ W$$

همچنین: $V_{out,DC} \simeq 0 \Rightarrow P_{load} = \frac{V_p^2}{2R_L} = 722.5\ mW$

$$\rightarrow \eta \simeq \frac{0.7225}{1.147} \simeq \%63$$

البته، ادامه خواهیم دید بازده واقعی مدار بیشتر از این مقدار تخمینی که ما محاسبه کردیم می‌باشد.

Introduction and Evaluation of the Output Stage

In this section, explain the following:

- How the output stage is connected to the preceding stages

  The output stage is connected to the preceding stages through a voltage buffer stage with an active load. The active load is an NMOS transistor that forms a current mirror with the transistor in the reference branch. In addition, transistor M12 is the only transistor in the circuit with a different $\frac{W}{L}$ (4 times that of the other transistors). This increases the swing by a few hundredths of a volt. The purpose of this buffer stage is to prevent the output stage from affecting the bias of the preceding stages and to isolate them.

- The type of complementary transistors used

  We have used a Darlington pair to provide sufficient current for the load. We have also included four diode-connected transistors (as diodes) so that the output-input characteristic of the output stage has no dead zone and no distortion occurs in the output waveform.

- The output-stage biasing method

  For the source and sink current sources of the output stage, we have formed current mirrors using two transistors, NMOS and PMOS, and the reference branch, thereby supplying the output-stage bias current. In the bias analysis section, we saw that this current is approximately $170\,\mu A$.

- Design considerations for applying feedback in the next step

  In feedback analysis, when we want to consider the load resistance as part of the circuit, R_load is placed in parallel with R_{2f}; therefore, we cannot use very small resistances as feedback resistors.

5 Applying Global Negative Feedback

Problem Requirements (Section 4)

After adding the output stage, apply global negative feedback to the complete circuit. Feedback must be taken from the final circuit output and returned to the differential input. The main feedback network must consist of only two resistors, R_f and R_g, and the closed-loop gain must be approximately 20.

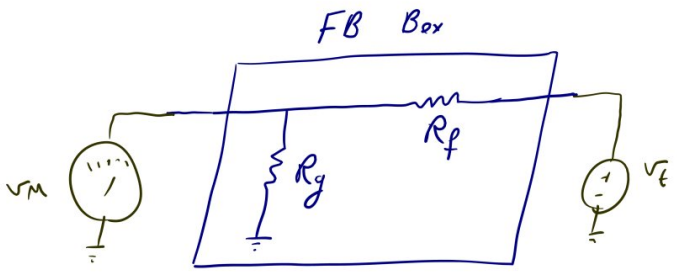
Theoretical Calculations of the Feedback Network

In this section:

- Write the closed-loop gain equations
- Select suitable values for R_f and R_g
- Verify that both resistances fall within the range of $10k\Omega$ to $200k\Omega$

$A_v = \frac{a'}{1+a'f} \xrightarrow{a'f \gg 1} A_v \simeq \frac{1}{f} \Rightarrow$ ی خواصیم: $f = \frac{1}{20}$

از یک فیدبک ولتاژ - ولتاژ به شکل زیر استفاده می‌کنیم:



$\rightsquigarrow f = \frac{R_g}{R_g + R_f} = \frac{1}{20} \rightsquigarrow \frac{R_g}{R_f} = 19 \rightsquigarrow R_f = 19 R_g$

$10^{k\Omega} < R_f, R_g < 200^{k\Omega} \rightarrow R_g = 10^{k\Omega}$

$R_f = 190^{k\Omega}$

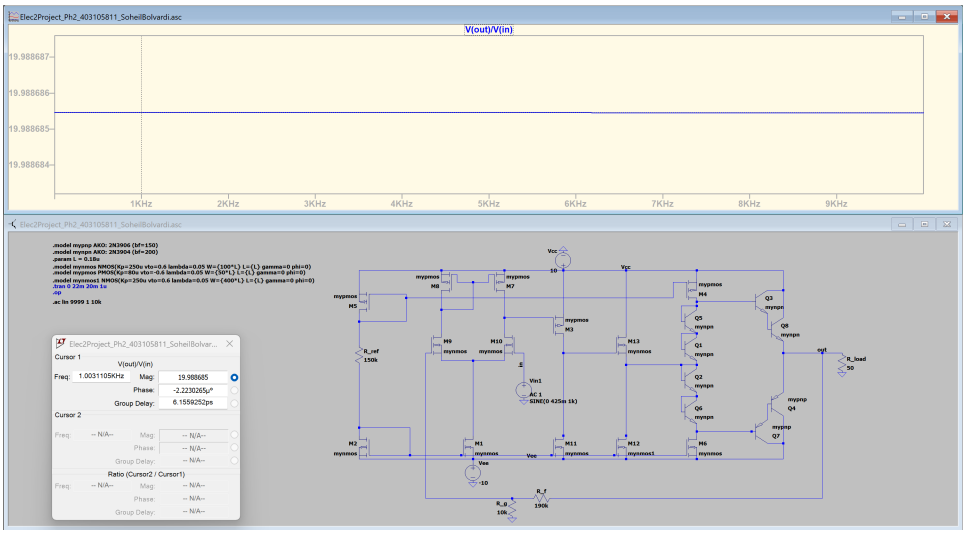
Results of Applying Feedback

Final values:

- R_f: $190\,k\Omega$
- R_g: $10\,k\Omega$
- Feedback sampling point: the final circuit output node (v_{out})
- Feedback application point: the positive terminal of the differential input (the side opposite the source)

Closed-loop gain at 1 kHz:

- $|A_{v,\text{closed}}|$: 19.989
- $20\log|A_{v,\text{closed}}|$: $26.016\,dB$



6 Comparison of Circuit Performance With and Without Feedback

Problem Requirements (Section 5)

Compare the complete circuit with and without feedback. The output stage must remain in the circuit in both cases. If the circuit saturates without feedback, explain how this affects the gain estimate.

Qualitative and Quantitative Comparison of the Two Cases

In this section, report the following:

- Comparison of the output waveforms in the two cases

  As shown in the image below, without feedback, the DC output value is approximately $-5.66\,V$, which severely limits the maximum swing to approximately $5.39\,V_{pp}$. After adding feedback, however, the DC output value reaches $2.55\,mV$ and the maximum swing reaches approximately $16.98\,V_{pp}$.

- Comparison of linearity

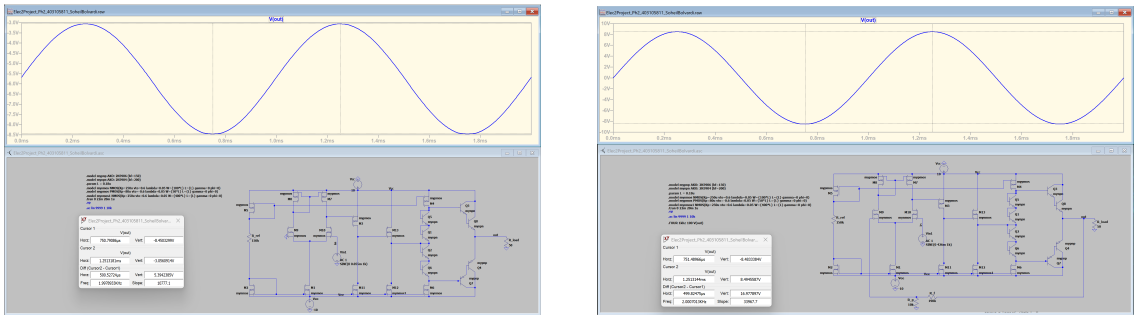
  The circuit without feedback is more linear, and we will demonstrate this quantitatively by comparing THD at the maximum swing for the cases with and without feedback. (The THD for the case with feedback is also examined separately in later sections.)

- Comparison of the onset of clipping

  Since the circuit gain without feedback is much larger (the open-loop gain, which is approximately 22000) and the swing is also more limited, clipping begins much earlier. The clipping threshold is approximately $0.055\,mV$ without feedback and approximately $426\,mV$ with feedback.

- Explanation of the effect of feedback on reducing distortion

  As stated, using feedback makes the circuit more linear. Therefore, we observe less distortion at the output. This reduction in distortion is not obvious enough to be identified from the appearance of the output waveform, but the significantly smaller THD with feedback shows that adding feedback makes the circuit more linear and reduces distortion.



Comparison of the output time-domain waveforms with feedback (left figure) and without feedback (right figure)

```
Partial Harmonic Distortion: 1.563457%
Total Harmonic Distortion:   1.563457%
```

```
Partial Harmonic Distortion: 0.076593%
Total Harmonic Distortion:   0.076663%
```

Comparison of THD at the maximum swing with feedback (left figure) and without feedback (right figure)

We observed that, after adding feedback, the circuit becomes more linear, its distortion decreases, and clipping occurs at a higher voltage.

7 Analysis of Output Swing, Power, and Efficiency

Problem Requirements (Section 6)

Determine whether the final circuit can produce, across a 50Ω load, a sinusoidal output with an amplitude of $16V_{pp}$ without severe clipping or visible distortion. Then measure the power delivered to the load, the total circuit power consumption, and the output-stage efficiency.

Theoretical Calculations of Swing and Efficiency

In this section:

- Estimate the maximum expected swing
- Calculate the load power using theoretical equations

در بخش‌های قبل نشان دادیم به دلیل استفاده از زوج دارلینگتون، جریان بار تأمین می‌شود.
بنابراین در بررسی سوئینگ باید خاموش/اشباع ترانزیستورهای طبقه خروجی را بررسی کنیم.

$$M4: OFF \rightsquigarrow V_{out,max} < 10 - 2V_{BE} = 8.6\,V$$

اما در واقع سوئینگ هیچ‌گاه نمی‌تواند به این مقدار برسد. زیرا قبل از خاموش شدن کامل ترانزیستور $M4$، وارد ناحیه triode شده و جریان آن به حدی کوچک می‌شود که جریان بار را تأمین نمی‌کند.
برای یافتن این آستانه:

با توجه به آینه جریان بدون $M4$ با نقطه مرجع و طبق بخش بایاس قرار می‌دهیم: $V_{SG,M4} = 0.85\,V$

$$\begin{cases} \dfrac{\beta_{npn}^2 \times I_{min}}{R_{out}} + 2V_{BE} = V_{OS,M4} \\ I_{min} = 0.08 \times 50 \times (0.85 - 0.6 - \frac{1}{2}V_{OS,M4}) \end{cases} \quad V_{OS,M4} \rightarrow M4: triode$$

که با حل دستگاه بالا خواهیم داشت $V^*_{OS,M4} \simeq 0.1\,V$

$$\rightsquigarrow V_{out,max} = 10 - V^*_{OS,M4} - 2V_{BE} \simeq \boxed{8.5\,V}$$

توان بار به ازای دامنه $8\,V$:

$$P_{load} = \frac{V_p^2}{2R_L} = \frac{8^2}{2\times 50} = \boxed{640\,mW}$$

همچنین در بخش‌های قبل دیدیم:

$$P_{supply} = 2V_{CC}\left(\frac{V_p}{\pi R_L} + I_{bias} + I_{\text{بقیه‌دار}}\right)$$

$$\rightsquigarrow V_p = 8\,V \simeq 1.045\,W \rightsquigarrow \eta \simeq \%61$$

البته در شبیه‌سازی می‌بینیم بازده مدار از این مقدار تئوری که محاسبه کردیم بیشتر است.

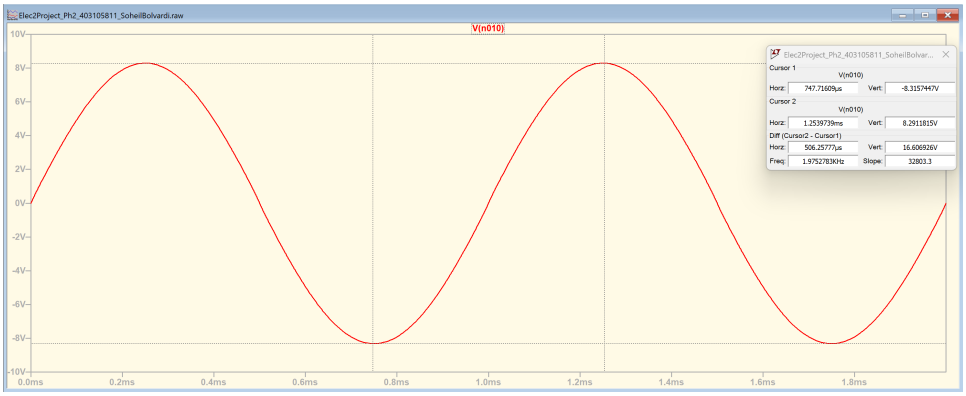
Transient Simulation and Power Measurement Results

In this section, include the following:
In this section, the plots and efficiency are presented for a swing of $16\,V_{pp}$, although the maximum swing of this circuit is approximately $17\,V_{pp}$. The plots, efficiency, and other specifications for the 17-volt swing are provided in the section on the project's bonus requirements (Section 15).

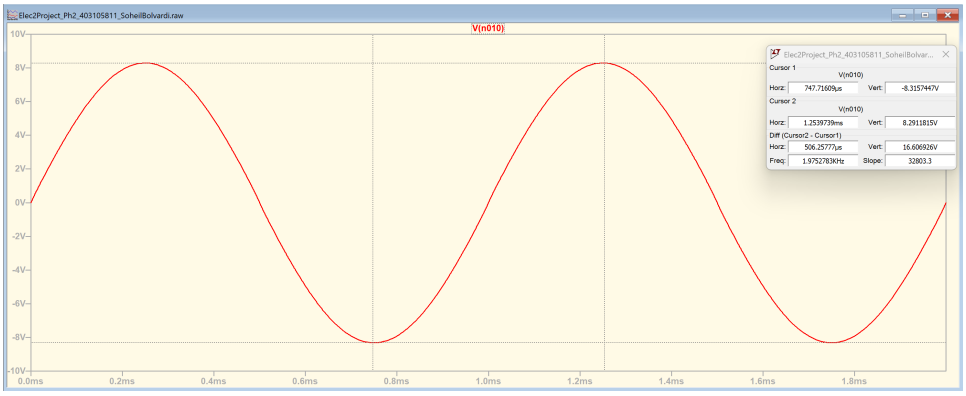
- Input amplitude required to achieve $16V_{pp}$

  A sinusoid with an amplitude of approximately $400.5\,V$

- Input waveform of the power stage

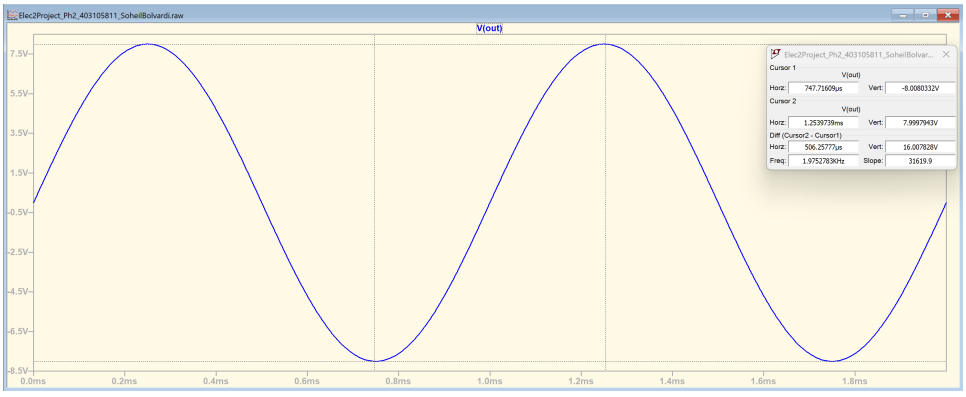


- Output waveform of the power stage



- Final circuit output waveform

  The final output is the same as the power-stage output.



- Examination of the effect of feedback on the onset of clipping

  This relates to the previous section and was explained there.

Numerical results: We used the following commands to obtain the load power, total power, and output-stage power, respectively:

```
.meas TRAN P_R_l RMS V(out)*I(R_load)
.meas TRAN P RMS abs(V(Vcc)*I(vcc))+abs(V(Vee)*I(Vee))
.meas TRAN P_OutputStage RMS abs(V(Vcc)*(Ic(Q3)+Ic(Q8)+Id(
    M4)))+abs(V(Vee)*(Id(M6)+Ic(Q4)+Ic(Q7)))
```

The SPICE simulation results are shown in the image below:

```
P_R_l: RMS(V(out)*I(R_load))=0.785002368001 FROM 0 TO 0.002
P: RMS(abs(V(Vcc)*I(vcc))+abs(V(Vee)*I(Vee)))=1.15770337559 FROM 0 TO 0.002
P_OutputStage: RMS(abs(V(Vcc)*(Ic(Q3)+Ic(Q8)+Id(M4)))+abs(V(Vee)*(Id(M6)+Ic(Q4)+Ic(Q7))))=1.13321275373 FROM 0 TO 0.002
```

Based on these results:

- Input amplitude required for $16V_{pp}$: $400.5\,V$
- P_{R_L}: $785.00\,mW$
- P_{total}: $1.15770\,W$
- $P_{\text{out-stage}}$: $1.13321\,W$
- η_{out}: $69.27\,\%$

$$\eta_{\text{out}} = \frac{P_{R_L}}{P_{\text{out-stage}}} \times 100\%$$

Commands used in the simulation:

```
.meas TRAN P_R_l RMS V(out)*I(R_load)
.meas TRAN P RMS V(Vcc)*I(vcc)-V(Vee)*I(vee)
```

8 Power Consumption at a Specified Input

Problem Requirements (Section 7)

For a sinusoidal input with an amplitude of $50mV$ and a frequency of $1\,\text{kHz}$, measure the total power consumption of the final circuit and compare it with the requirement

$$P_{\text{total}} \leq 190mW$$

given above.

Power Consumption Measurement Under Standard Conditions

In this section, report the result of measuring the total circuit power under the project's standard conditions.

Parameter	Value
Input amplitude	$50mV$
Input frequency	$1\,\text{kHz}$
Total circuit power consumption	$175.48\,mV$
Pass status	Pass

We used the same command as in the previous section to calculate the total circuit power:

```
.meas TRAN P RMS abs(V(Vcc)*I(vcc))+abs(V(Vee)*I(Vee))
```

The result is shown below:

```
P_R_l: RMS(V(out)*I(R_load))=0.0122346169425 FROM 0 TO 0.002
P: RMS(abs(V(Vcc)*I(vcc))+abs(V(Vee)*I(Vee)))=0.175476678923 FROM 0 TO 0.002
```

9 Measurement of Total Harmonic Distortion THD

Problem Requirements (Section 8)

Measure THD for the final circuit with feedback, including the output stage and the 50Ω load. The test must be performed using a sinusoidal input with an amplitude of $50mV$ and a frequency of 1 kHz. Then repeat the same test with a noise source connected in series with the resistors that set the currents of the current sources.

Measurement Method and Analysis of Results

In this section, explain the test method and, if necessary, describe:

- How steady state was reached

  In the circuit with feedback, the output usually exhibits transient distortion during the initial cycles. Therefore, in all time-domain simulations in this report, we performed the simulation over the interval $20 - 22\,ms$.

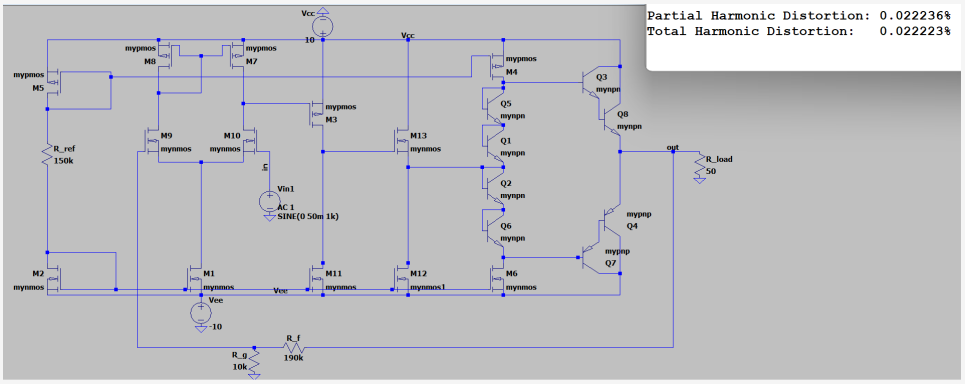
- The points at which the noise source was connected in series It is connected in series with the reference-branch resistor, which is the only resistor in the circuit. It is also connected in series with the PMOS transistor of the amplification stage, changing its gate-source voltage; this noise changes the transistor's output current and voltage.

- Why THD is expected to increase in the presence of noise As noise increases, changes in the transistor voltages and currents make the circuit more nonlinear, and THD increases because of the resulting distortion in the sinusoidal output waveform.

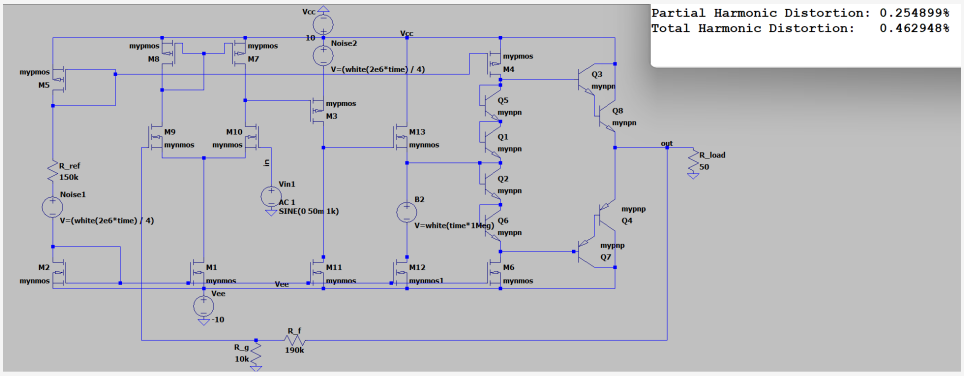
Suggested command:

```
.FOUR 1khz 100 V(out)
```

The circuit diagram and THD under ideal conditions:



The circuit diagram showing the locations of the noise sources, and the THD when noise is present in the circuit:



THD Test Results

Test conditions	THD value	Status
Circuit with feedback, without a noise source	0.022%	Must be less than 0.08%
Circuit with feedback, with a series noise source	0.463%	Must be less than 1%

Brief explanation of the change in THD in the presence of noise: As stated, as noise increases, changes in the transistor voltages and currents make the circuit more nonlinear, and THD increases because of the resulting distortion in the sinusoidal output waveform.

10 Measurement of PSRR With a Sawtooth Power Supply

Problem Requirements (Section 9)

Set the circuit input to zero and examine the effect of power-supply ripple on the output. In this test, sawtooth power supplies with a ripple of $10mV$ must be used for V_{CC} and V_{EE}, and the value of $PSRR$ must then be calculated using the equation

$$PSRR = 20 \log \left(\frac{V_{\text{ripple}}}{\Delta V_{out}} \right)$$

shown above.

$PSRR$ Test Method and Calculation

In this section:

- Specify how the ripple is defined (peak or peak-to-peak)

  We define the ripple as peak-to-peak for a reference voltage of $10\,V$. The sources used and the waveforms of V_{CC} and V_{EE} are provided below.

```
Vcc Vcc 0 PWL(20m 10 25m 10.01 30m 10 35m
    10.01 40m 10)
Vee Vee 0 PWL(20m -10 25m -10.01 30m -10 35m
    -10.01 40m -10)
```

- Explain the method for measuring ΔV_{out}

  To measure the effect of power-supply ripple on the output ripple, we set the circuit input to zero and plot the output in the time domain. The output waveform is shown below.

- Describe how $PSRR$ is calculated

  To calculate PSRR, we use the same equation stated earlier, namely

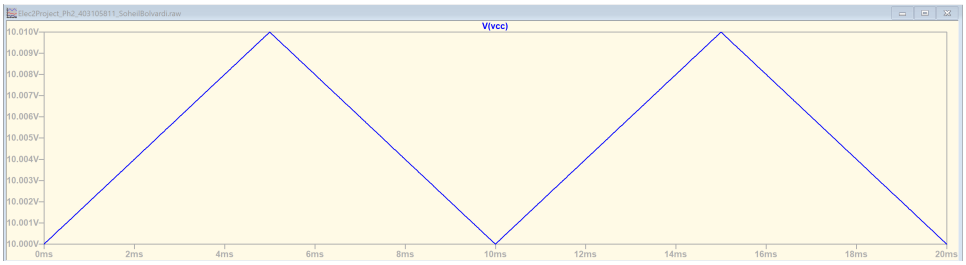
  $$PSRR = 20 \log \left(\frac{V_{\text{ripple}}}{\Delta V_{out}} \right)$$

  as shown above.

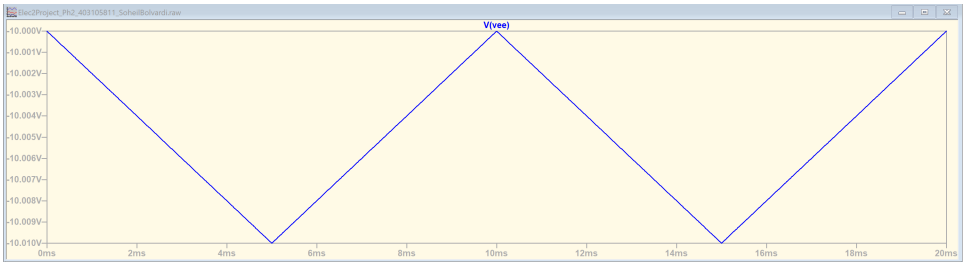
Suggested example source:

```
Vcc Vcc 0 PWL(0 12 5m 12.01 10m 12 15m 12.01 20m 12)
Vee Vee 0 PWL(0 -12 5m -12.01 10m -12 15m -12.01 20m -12)
```

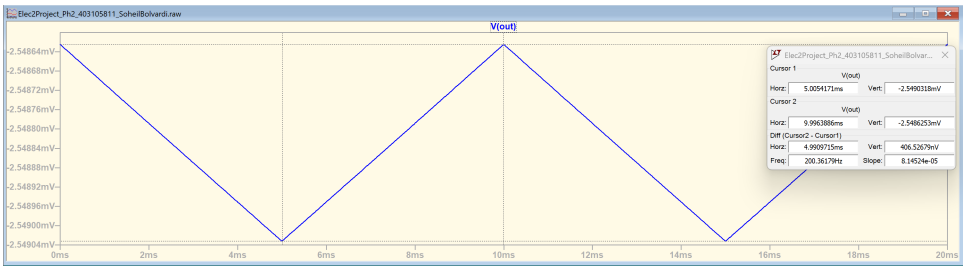
$PSRR$ Test Results



Waveform of V_{CC}



Waveform of V_{EE}



Waveform of $V(out)$

Parameter	Value
V_{ripple}	$10\,mV$
ΔV_{out}	$406\,nV$
$PSRR$	$87.83\,dB$

Brief explanation of the circuit's performance against power-supply ripple:
PSRR, or Power Supply Rejection Ratio, is the ratio that quantifies the rejection of power-supply fluctuations. This factor indicates the extent to which power-supply noise and ripple affect the output. Since feedback is used in the circuit, we observe that the circuit performs very well against supply noise.

11 Measurement of Differential Input Resistance

Problem Requirements (Section 10)

Measure the differential input resistance of the final circuit and compare it with the requirement

$$R_{in,\text{diff}} > 1M\Omega$$

given above. The measurement method must be fully explained.

Calculations or Analytical Method for Differential Input Resistance

In this section, consider the equivalent input model and, if possible, provide a theoretical estimate of the input resistance.

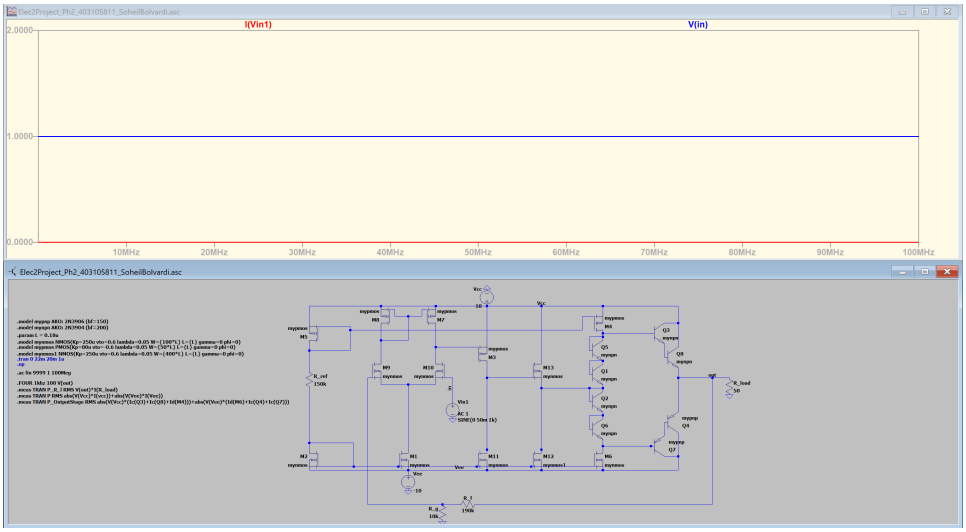
In the differential stage of this circuit, MOS transistors are used, and the input voltage is applied to the transistor gate. Since the gate draws no current, the input resistance is ∞, because the gate current is exactly zero. We also demonstrate this in the simulation section that follows.

Differential Input Resistance Simulation Results

In this section, explain the exact measurement method. For example:

- The type of test source
- How differential excitation is applied
- The equation for obtaining resistance from the voltage-to-current ratio

To simulate this section, we place a source at the input, perform an AC analysis, and read its voltage-to-current ratio. By superposition, this ratio is the differential input resistance.



We observe that, at all frequencies, the current of the input source (the test source) is exactly zero. Therefore, the input resistance is ∞. Naturally, because the current is zero, nothing is displayed if we plot the voltage-to-current ratio directly.

Final value:

- $R_{in,\text{diff}}$: ∞
- Pass status: Pass

12 Measurement of Output Resistance

Problem Requirements (Section 11)

Measure the output resistance of the final circuit after removing the load R_L and compare it with the requirement

$$R_{out} < 50\Omega$$

given above. The simulation method and the procedure for extracting the value must be explained.

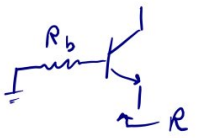
Calculations or Analytical Method for Output Resistance

In this section, describe the theoretical method or analytical estimate of the output resistance.

← همانگونه که گفته شد ، در طبقه فرودی از زوج دارلینگتون استفاده کرده ایم. می دانیم که برای زوج دارلینگتون :

$$\beta_{eq} = \beta_1 \beta_2$$

و همچنین :



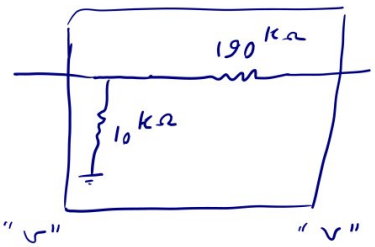
$$R = \frac{1}{g_m} + \frac{R_b}{\beta}$$

← بنابراین با توجه به بزرگ بودن β_{eq} ، از تأثیر مقاومت کم بیس صرف نظر می کنیم.



$$g_{m_{eq}} = \beta_{npn}\, g_{m_2} \rightsquigarrow \begin{cases} g_{m,eq,npn} \simeq 92.66\ m\mho \\ g_{m,eq,pnp} \simeq 92.52\ m\mho \end{cases}$$

← شبکه فیدبک



$$\rightsquigarrow r_{2f} = 190^{k\Omega} + 10^{k\Omega} = 200^{k\Omega}$$

پس در نهایت مقاومت خروجی با وجود بار، برابر است با :

$$r'_o = \frac{1}{g_{m,npn}} \parallel \frac{1}{g_{m,pnp}} \parallel 200^{k\Omega} \parallel 50^{\Omega} \simeq 4.90^{\Omega}$$

همچنین ، از شبیه سازی می دانیم :

Open loop gain : $A_v \simeq 20000$

$$\rightsquigarrow 1 + af \simeq 1001$$

$$R'_{out} = \frac{r'_o}{1+af} = \frac{4.90^{\Omega}}{1001} \simeq 4.89^{m\Omega}$$

$$R'_{out} = R_{load} \parallel R_{out} \rightarrow R_{out} \parallel 50^{\Omega} = 4.89^{m\Omega}$$

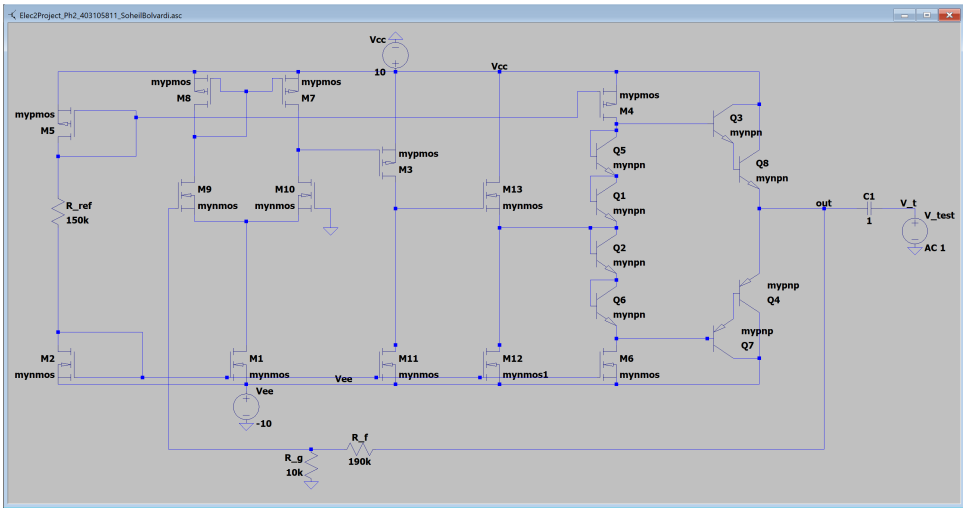
$$\rightsquigarrow R_{out} \simeq R'_{out} \rightarrow \boxed{R_{out} \simeq 4.89^{m\Omega}}$$

Output Resistance Simulation Results

In this section, explain the simulation method. For example:

- Setting the input to zero
- Removing the output load
- Applying a test source at the output
- Obtaining resistance from the voltage-to-current ratio

The simulation procedure is to first set the input to zero and remove the circuit's load resistor. We then place a test voltage source at the output. A capacitor with a capacitance of $1\,F$ is placed between this source and the circuit because the test source must not change the circuit's DC conditions. We choose a large capacitance so that its effect at low frequencies quickly becomes negligible, allowing us to observe the circuit's own characteristic. Finally, we read the test source's voltage-to-current ratio, which is the output resistance. The circuit used to obtain the output resistance is shown below:





Final value:

- R_{out}: $4.90\,m\Omega$
- Pass status: Pass

13 Real Audio Signal Amplification Test

Problem Requirements (Section 12)

In this section, a real audio file must be applied to the circuit input, and the amplified output must be examined. Using a simple sinusoidal signal instead of a real audio file is not acceptable. The input and output waveforms must be included, and it must be shown that the output does not exhibit severe clipping.

Audio File Preparation and Necessary Explanations

In this section, specify:

- The type of input file
- Whether it was converted to a single channel
- The sampling rate
- Whether the amplitude was normalized

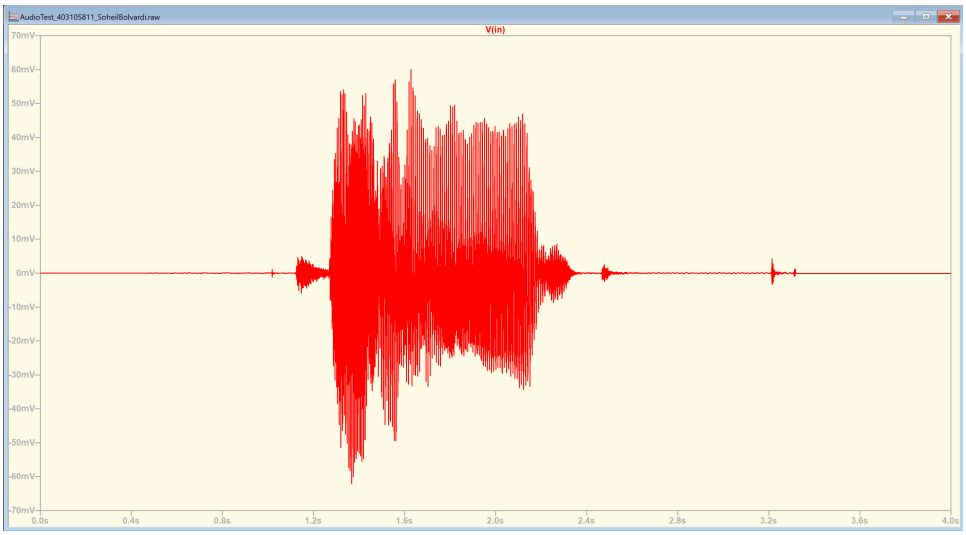
For the input file, we use the same sample file provided in the template. We run the supplied script as shown below to generate the audio.wav file. This file is single-channel, and after checking it in MATLAB, we found that the sampling rate is the standard rate of $44.1\,kHz$. The function used also performs normalization itself:



Audio Test Results

Audio file specifications:

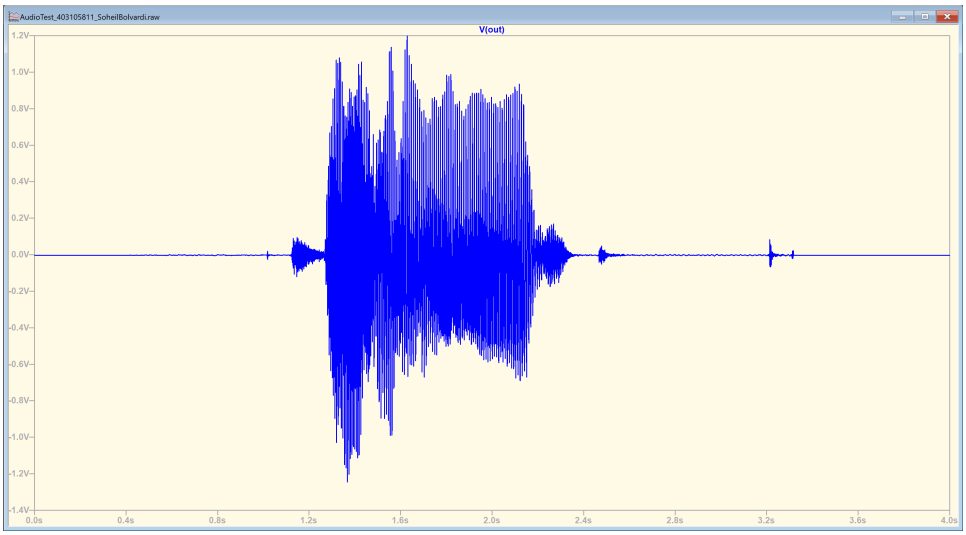
- Input filename: $audio.wav$
- Output filename: $output.wav$
- fft of the input and output files



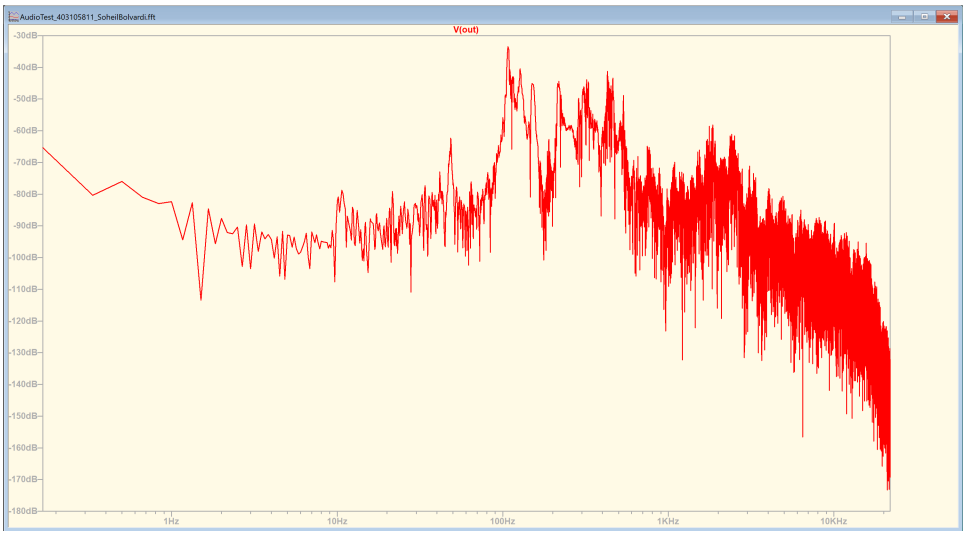
Input waveform



Input fft

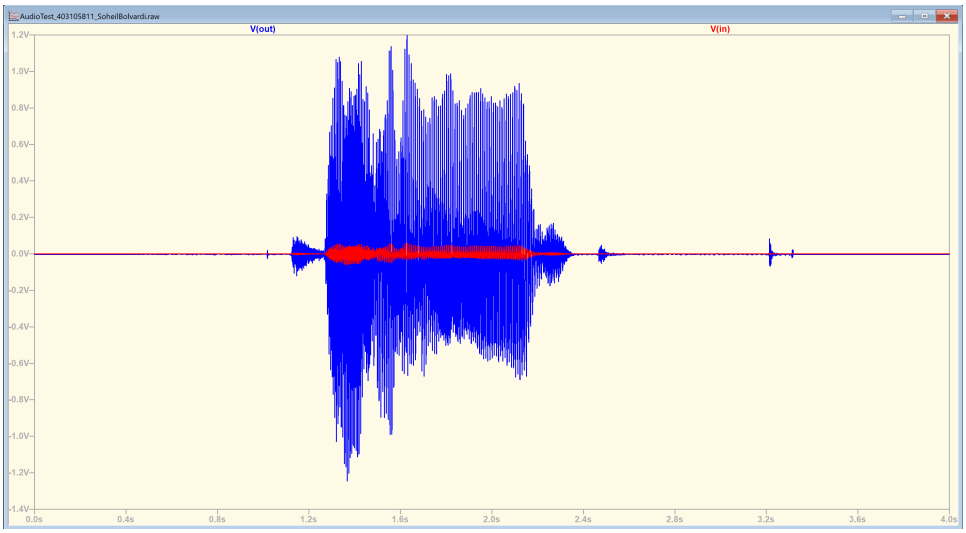


Output waveform



Output fft

Explanation of output quality and the presence or absence of clipping:
We observe that the input and output waveforms have the same shape and that no clipping has occurred. In the image below, the input and output waveforms are shown simultaneously:



Comparison of the input and output waveforms

In the images above, we also saw that the Fourier transforms of the input and output have exactly the same shape and differ only by a constant factor. Therefore, we can conclude that the waveform has not changed and has only been amplified. In other words, no clipping has occurred.

Files submitted for this section:

- audio.wav
- output.wav

14 Circuit Cost and Component Constraints

Problem Requirements (Section 13)

Determine the final cost of your circuit using the cost calculation file. If the cost exceeds 200, calculate the penalty using the following equation:

$$\text{Penalty} = 2\% \times \left\lceil \frac{\text{Cost} - 200}{5} \right\rceil$$

Component Count and Cost Table

Component type	Quantity	Unit cost	Total cost
MOSFET	13	3	39
BJT	8	5	40
Resistor	3	10	30
Capacitor	0	40	0
Total cost			109

Results:

- Total circuit cost: 109
- Have you exceeded the cost limit? No
- Penalty amount, if any: 0

Using the supplied Python script to calculate the cost, we obtain:

```
Elec2Project_Ph2_403105811_SoheilBolvardi.asc: 109.0
```

Total circuit cost

15 Project Bonus Requirements

Problem Requirements (Section 14)

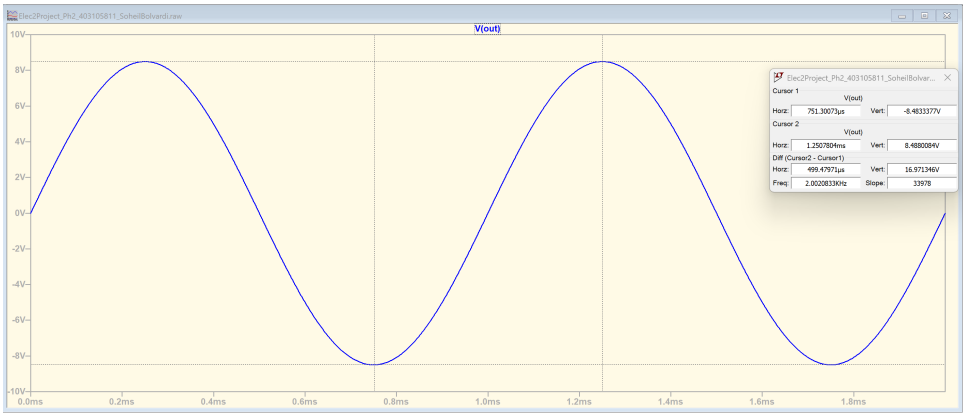
If you have examined the project's bonus requirements, report the results in this section. To receive extra credit, all three of the following conditions must be met simultaneously:

- An output swing of $17V_{pp}$ without visible clipping
- Efficiency greater than 70% at the same operating point
- DC output voltage below $100mV$

Evaluation of the Bonus Requirements

Parameter	Requirement	Your value
Output swing	$17V_{pp}$	$16.97V_{pp}$
Efficiency	$> 70\%$	72.06%
$\lvert V_{out,DC} \rvert$	$< 100mV$	$2.55\,mV$

Regarding the swing value, the project staff were consulted and confirmed that a swing of $16.97\,V_{pp}$ is accepted for the bonus section.



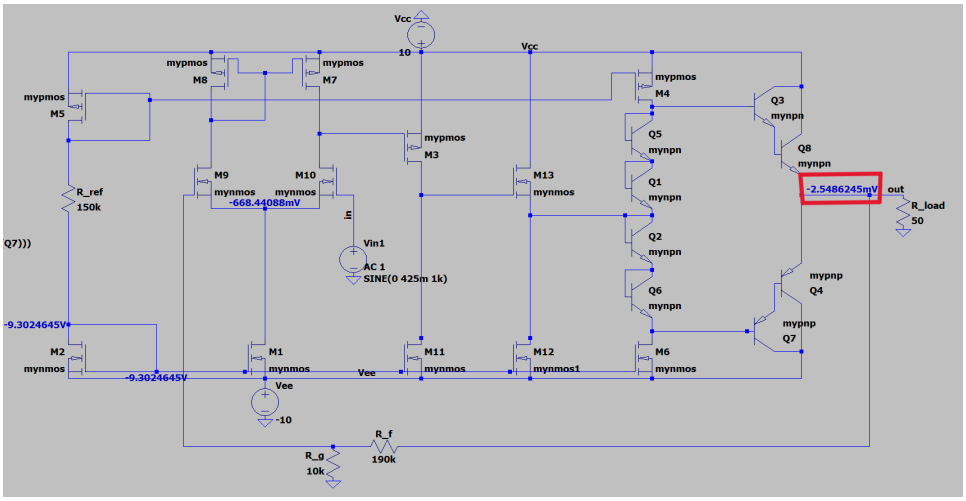
Output swing of $16.97\,V_{pp}$

We observe that there is no distortion in the output waveform.

```
P_R_1: RMS(V(out)*I(R_load))=0.883826932122 FROM 0 TO 0.002
P: RMS(abs(V(Vcc)*I(vcc))+abs(V(Vee)*I(Vee)))=1.2265173661 FROM 0 TO 0.002
```

Circuit efficiency of 72.06% at a swing of $16.97\,V_{pp}$

In the bias calculation section, we also saw that the circuit's DC output value is $2.55\,mV$:



DC output voltage value

Required items:

- Input amplitude corresponding to $17V_{pp}$: $425.5\,mV$
- Output waveform under this condition: attached above
- Final result of the bonus section: Pass

16 Submitted Files

Problem Requirements (Section 15)

Complete the list of final files submitted for the project.

Submitted Files Checklist

- The complete LTspice circuit schematic file with the .asc extension
- The file related to the audio test
- The audio.wav file
- The output.wav file
- The final report in PDF format

17 Final Summary

Problem Requirements (Section 16)

In a brief summary, explain:

- What was the main design challenge in phase two?
- Which part was more difficult: the output stage, feedback, THD, $PSRR$, or the audio test?
- If some specifications were not fully met, what was the reason?
- What improvements would you make to the circuit if you had more time or a larger budget?

Final Conclusion

The main design challenge in phase two was increasing the swing while keeping the total circuit power consumption low. The output stage was more difficult than the other parts mentioned, although it did not present any major difficulty. Fortunately, all the required specifications were met, and there were no issues in this respect. One aspect of this project that could be improved is the total circuit power consumption, since it was closer to the specified limit than the other specifications. The THD could also have been lower than the value obtained here (namely 0.022%).